ADI schemes for pricing options under the Heston model

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Abstract

In this note, we present the theory of the Heston model and the Alternating Direction Implicit (ADI) method, following [HF10]. A key contribution of this note is the replacement of Dirichlet boundary conditions at $s = 0$ and $v = V_{\max}$ with time-derivative conditions, which avoids compatibility issues at $t = 0$ and simplifies implementation by treating all grid points uniformly.

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1 Heston model

European-style call option gives the holder the right to buy a given asset at a prescribed *maturity date* T for a prescribed *strike price* K.

Let S_t denote the asset price at time $t \geq 0$. The *payoff* of the call option is $\phi(S_T)$ with $\phi(s) = (s-K)^+ := \max\{0, s-K\}$.

The Heston stochastic volatility model [Hes93] is given by the stochastic differential equation

$$\begin{cases} \mathrm{d}S_t & = (r_d - r_f)S_t \,\mathrm{d}t + \sqrt{V_t} S_t \,\mathrm{d}W_t^1, \\ \mathrm{d}V_t & = \kappa(\eta - V_t)\,\mathrm{d}t + \sigma\sqrt{V_t}\,\mathrm{d}W_t^2, \\ \mathrm{d}\langle W^1, W^2\rangle_t & = \rho\,\mathrm{d}t, \end{cases}$$

where

- $\rho \in [-1, 1]$ is the correlation factor,
- $\kappa \geq 0$ mean-reversion rate,
- $\sigma \geq 0$ volatility-of-variance,
- $r_d \geq 0$ domestic interest rate, and
- $r_f \geq 0$ foreign interest rate.

Let

$$u(t, s, v) := \mathbb{E}[e^{-r_d t}\phi(S_T) | (S_{T-t}, V_{T-t}) = (s, v)].$$

Then $u(t, s, v)$ is the fair value of a European call option if $(S_{T-t}, V_{T-t}) = (s, v)$, where t is time till maturity. Let $X_t := (S_t, V_t)$. Then its quadratic variation is

$$\mathrm{d}\langle X\rangle_t = \begin{bmatrix} \sqrt{V_t}S_t & 0 \\ 0 & \sigma\sqrt{V_t} \end{bmatrix} \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \begin{bmatrix} \sqrt{V_t}S_t & 0 \\ 0 & \sigma\sqrt{V_t} \end{bmatrix}^\top \mathrm{d}t \tag{1.1}$$

$$= \begin{bmatrix} V_t S_t^2 & \rho\sigma V_t S_t \\ \rho\sigma V_t S_t & \sigma^2 V_t \end{bmatrix} \mathrm{d}t. \tag{1.2}$$

Let

$$\alpha(t, s, v) = \begin{bmatrix} (r_d - r_f)s \\ \kappa(\eta - v) \end{bmatrix}, \tag{1.3}$$

$$\beta(t, s, v) = \begin{bmatrix} vs^2 & \rho\sigma vs \\ \rho\sigma vs & \sigma^2 v \end{bmatrix}. \tag{1.4}$$

By Feynman–Kac formula, $u(t,s,v)$ satisfies the PDE

$$\begin{cases} \partial_t u & = -r_d u + \alpha \cdot \nabla_x u + \frac{1}{2}\beta : \nabla_x^2 u, \\ u(0,s,v) & = \phi(s). \end{cases} \tag{1.5}$$

A boundary condition at $s = 0$ holds:

$$u(t,0,v) = 0. \tag{1.6}$$

By [ET10], the PDE is satisfied at $v = 0$.

2 Discretization scheme

Our aim is to construct a stable, efficient and robust numerical discretization scheme for (1.5). Fix $S_{\max} > 0$ and $V_{\max} > 0$ large enough. Then we work on the bounded domain $[0,T] \times [0, S_{\max}] \times [0, V_{\max}]$. So we need additional boundary conditions:

$$\partial_s u(t, S_{\max}, v) = e^{-r_f t}, \tag{2.1}$$

$$u(t, s, V_{\max}) = s e^{-r_f t}. \tag{2.2}$$

Heuristically, we choose

$$S_{\max} = 30K \quad \text{and} \quad V_{\max} = 15.$$

For numerical and financial reasons, we need relatively many mesh points near $s = K$ and $v = 0$. We will use the hyperbolic sine

$$\sinh(\xi) = \frac{e^{\xi} - e^{-\xi}}{2}.$$

2.1 Mesh in s-direction

Let $m_1 \geq 2$ be an integer, $c > 0$ a constant, and $0 < S_{\text{left}} < K < S_{\text{right}} < S_{\max}$. Let equidistant points $\xi_{\min} = \xi_0 < \xi_1 < \cdots < \xi_{m_1} = \xi_{\max}$ be given by

$$\xi_{\min} = \sinh^{-1}\Big(-\frac{S_{\text{left}}}{c}\Big), \tag{2.3}$$

$$\xi_{\text{int}} = \frac{S_{\text{right}} - S_{\text{left}}}{c}, \tag{2.4}$$

$$\xi_{\max} = \xi_{\text{int}} + \sinh^{-1}\Big(\frac{S_{\max} - S_{\text{right}}}{c}\Big). \tag{2.5}$$

Note that $\xi_{\min} < 0 < \xi_{\text{int}} < \xi_{\max}$. The nonuniform mesh

$$0 = s_0 < s_1 < \cdots < s_{m_1} = S_{\max}$$

of $[0, S_{\max}]$ is defined by the transformation

$$s_i = \varphi(\xi_i) \quad \text{for} \quad 0 \le i \le m_1,$$

where

$$\varphi(\xi) = \begin{cases} S_{\text{left}} + c\sinh(\xi) & \text{if} \quad \xi \in [\xi_{\min}, 0], \\ S_{\text{left}} + c\xi & \text{if} \quad \xi \in (0, \xi_{\text{int}}), \\ S_{\text{right}} + c\sinh(\xi - \xi_{\text{int}}) & \text{if} \quad \xi \in [\xi_{\text{int}}, \xi_{\max}]. \end{cases}$$

Then the mesh is uniform inside $[S_{\text{left}}, S_{\text{right}}]$ and nonuniform outside. Heuristically, we choose

$$c = \frac{K}{10}, \tag{2.6}$$

$$S_{\text{left}} = K \max\left\{\frac{1}{2}, e^{-rT}\right\}, \tag{2.7}$$

$$S_{\text{right}} = K \min\left\{\frac{3}{2}, e^{rT}\right\}, \tag{2.8}$$

where $r = \frac{1}{10}$.

2.2 Mesh in v-direction

Let $m_2 \ge 2$ be an integer and $d > 0$ a constant. Let equidistant points $\psi_0 < \psi_1 < \cdots < \psi_{m_2}$ be given by

$$\psi_j = j\Delta\psi \quad \text{for} \quad 0 \le j \le m_2,$$

where

$$\Delta\psi = \frac{1}{m_2}\sinh^{-1}\left(\frac{V_{\max}}{d}\right).$$

The nonuniform mesh

$$0 = v_0 < v_1 < \cdots < v_{m_2} = V_{\max}$$

of $[0, V_{\max}]$ is defined by the transformation

$$v_j = d\sinh(\psi_j) \quad \text{for} \quad 0 \le j \le m_2.$$

Heuristically, we choose

$$d = \frac{V_{\max}}{500}.$$

A key departure from the standard approach (see e.g. 't Hout's slides[1]) occurs in Section 5, where we replace the Dirichlet conditions at $s = 0$ and $v = V_{\max}$ with time-derivative conditions that are more compatible with the initial data and easier to implement.

3 Finite difference approximation

Let $f : \mathbb{R} \to \mathbb{R}$ and discrete points $\cdots < x_{i-1} < x_i < x_{i+1} < \cdots$. We denote

$$\Delta_i \coloneqq x_i - x_{i-1}, \tag{3.1}$$
$$f_i \coloneqq f(x_i), \tag{3.2}$$
$$f_i' \coloneqq f'(x_i), \tag{3.3}$$
$$f_i'' \coloneqq f''(x_i). \tag{3.4}$$

For first derivative,

$$\text{backward:} \quad f_i' \approx \alpha_{-2} f_{i-2} + \alpha_{-1} f_{i-1} + \alpha_0 f_i, \tag{3.5}$$
$$\text{central:} \quad f_i' \approx \beta_{-1} f_{i-1} + \beta_0 f_i + \beta_1 f_{i+1}, \tag{3.6}$$
$$\text{forward:} \quad f_i' \approx \gamma_0 f_i + \gamma_1 f_{i+1} + \gamma_2 f_{i+2}, \tag{3.7}$$

where the coefficients are

$$\alpha_{-2} = \frac{\Delta_i}{\Delta_{i-1}(\Delta_{i-1} + \Delta_i)}, \qquad \alpha_{-1} = \frac{-\Delta_{i-1} - \Delta_i}{\Delta_{i-1}\Delta_i}, \qquad \alpha_0 = \frac{\Delta_{i-1} + 2\Delta_i}{\Delta_i(\Delta_{i-1} + \Delta_i)},$$
$$\beta_{-1} = \frac{-\Delta_{i+1}}{\Delta_i(\Delta_i + \Delta_{i+1})}, \qquad \beta_0 = \frac{\Delta_{i+1} - \Delta_i}{\Delta_i\Delta_{i+1}}, \qquad \beta_1 = \frac{\Delta_i}{\Delta_{i+1}(\Delta_i + \Delta_{i+1})},$$
$$\gamma_0 = \frac{-2\Delta_{i+1} - \Delta_{i+2}}{\Delta_{i+1}(\Delta_{i+1} + \Delta_{i+2})}, \qquad \gamma_1 = \frac{\Delta_{i+1} + \Delta_{i+2}}{\Delta_{i+1}\Delta_{i+2}}, \qquad \gamma_2 = \frac{-\Delta_{i+1}}{\Delta_{i+2}(\Delta_{i+1} + \Delta_{i+2})}. \tag{3.8}$$

(3.9)

(3.10)

[1] http://quantshub.com/qhworkshopview/22

For second derivative,

$$\text{backward:} \quad f_i'' \approx \lambda_{-2} f_{i-2} + \lambda_{-1} f_{i-1} + \lambda_0 f_i, \tag{3.11}$$
$$\text{central:} \quad f_i'' \approx \delta_{-1} f_{i-1} + \delta_0 f_i + \delta_1 f_{i+1}, \tag{3.12}$$
$$\text{forward:} \quad f_i'' \approx \kappa_0 f_i + \kappa_1 f_{i+1} + \kappa_2 f_{i+2}, \tag{3.13}$$

where

$$\begin{aligned}
\lambda_{-2} &= \frac{2}{\Delta_{i-1}(\Delta_{i-1}+\Delta_i)}, & \lambda_{-1} &= \frac{-2}{\Delta_{i-1}\Delta_i}, & \lambda_0 &= \frac{2}{\Delta_i(\Delta_{i-1}+\Delta_i)}, \\
\delta_{-1} &= \frac{2}{\Delta_i(\Delta_i+\Delta_{i+1})}, & \delta_0 &= \frac{-2}{\Delta_i\Delta_{i+1}}, & \delta_1 &= \frac{2}{\Delta_{i+1}(\Delta_i+\Delta_{i+1})}, \\
\kappa_0 &= \frac{2}{\Delta_{i+1}(\Delta_{i+1}+\Delta_{i+2})}, & \kappa_1 &= \frac{-2}{\Delta_{i+1}\Delta_{i+2}}, & \kappa_2 &= \frac{2}{\Delta_{i+2}(\Delta_{i+1}+\Delta_{i+2})}.
\end{aligned} \tag{3.14}$$

(3.15)

(3.16)

Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ and discrete points $\cdots < x_{i-1} < x_i < x_{i+1} < \cdots$ and $\cdots < y_{j-1} < y_j < y_{j+1} < \cdots$. We denote

$$f_{ij} \coloneqq f(x_i, y_j), \tag{3.17}$$
$$f_{ij}'' \coloneqq \partial_x \partial_y f(x_i, y_j). \tag{3.18}$$

For mixed derivatives,

$$\text{central:} \quad f_{ij}'' \approx \sum_{k,l=-1}^{1} \beta_k^x \beta_l^y f_{i+k,j+l}.$$

4 Computational grid

The grid over (s, v) is

$$\mathcal{G} \coloneqq \{(s_i, v_j)\colon 0 \le i \le m_1, 0 \le j \le m_2\}.$$

If $i \le m_1 - 1$:

- $\partial_s u(t,s,v)$: central everywhere,
- $\partial_v u(t,s,v)$: forward at $v_0 = 0$, central if $0 < v_j < 1$, backward if $v_j \geq 1$,
- $\partial^2_{ss} u(t,s,v), \partial^2_{vv} u(t,s,v), \partial^2_{sv} u(t,s,v)$: central everywhere.

If $i = m_1$:

- $\partial_s u(t,s,v)$: given by Neumann boundary condition (2.1),
- $\partial_v u(t,s,v)$: as above,
- $\partial^2_{ss} u(t,s,v)$: see below,
- $\partial^2_{vv} u(t,s,v)$: as above,
- $\partial^2_{sv} u(t,s,v)$: always zero.

Consider

$$f''(x) \approx \frac{f(x-h) - 2f(x) + f(x+h)}{h^2}.$$

If $f'(x)$ is given and $x+h$ outside the domain, then replace $f(x+h)$ by $f(x-h) + 2hf'(x)$. In this case, we obtain

$$f''(x) \approx \frac{2f(x-h) - 2f(x) + 2hf'(x)}{h^2}.$$

Semi-discretization leads to IVP for a system of $m = (m_1+1)(m_2+1)$ ODEs:

$$\begin{cases} U'(t) &= AU(t) + g(t) \quad \text{for} \quad t \in [0,T], \\ U(0) &= U_0, \end{cases}$$

where A is a given matrix and $g(t)$ a given vector.

Vector $U(t)$ approximates function $u(t,\cdot,\cdot)$ on $\mathcal{G}$. Vector U_0 is given by direct evaluation of the payoff function on $\mathcal{G}$ with one modification: the grid point s_i nearest to K take the cell average value

$$\frac{1}{h}\int_{s_{i-1/2}}^{s_{i+1/2}} \max\{0, s-K\}\,\mathrm{d}s,$$

where

$$s_{i-1/2} = \frac{s_{i-1}+s_i}{2}, \tag{4.1}$$

$$s_{i+1/2} = \frac{s_i+s_{i+1}}{2}, \tag{4.2}$$

$$h = s_{i+1/2} - s_{i-1/2}. \tag{4.3}$$

The next section is about obtaining A and g given a 2D second-order linear PDE.

5 From matrix ODE to vector ODE

The standard approach, as in the implementation of in 't Hout (slides[2] 32–42), restricts the computational domain to the interior points, and handles the boundary values $u(t, s, V_{\max}) = se^{-r_f t}$ separately via the forcing matrix. Here we instead keep all grid points — boundary and interior — as unknowns, and replace the Dirichlet conditions (5.4) and (5.5) with the time-derivative conditions (5.8) and (5.9), which are then incorporated directly into the system.

We consider the second-order linear PDE

$$u_t = \alpha_0 u + \alpha_1 u_s + \alpha_2 u_v + \beta_{11} u_{ss} + 2\beta_{12} u_{sv} + \beta_{22} u_{vv}, \tag{5.1}$$

with $(s, v) \in [0, S_{\max}] \times [0, V_{\max}]$.

We impose the initial and boundary conditions

$$u(0, s, v) = \phi(s), \tag{5.2}$$
$$\partial_s u(t, S_{\max}, v) = e^{-r_f t}, \tag{5.3}$$
$$u(t, 0, v) = 0, \tag{5.4}$$
$$u(t, s, V_{\max}) = se^{-r_f t}. \tag{5.5}$$

These conditions are "almost" equivalent to

$$u(0, s, v) = \phi(s), \tag{5.6}$$
$$\partial_s u(t, S_{\max}, v) = e^{-r_f t}, \tag{5.7}$$
$$\partial_t u(t, 0, v) = 0, \tag{5.8}$$
$$\partial_t u(t, s, V_{\max}) = -r_f s e^{-r_f t}. \tag{5.9}$$

The conditions (5.8) and (5.9) are preferable to (5.4) and (5.5) in the following two respects:

1. Mathematics: (5.2) and (5.5) are conflicting at $t = 0$ and $s > K$ where K is the strike price. On the other hand, (5.9) only governs how the function grows in time. So (5.9) is more compatible with (5.2).

[2] http://quantshub.com/qhworkshopview/22

2. Implementation: using (5.5) requires us to separate the interior unknown values from the known values at the boundary. This would complicate the Python coding. The advantage of (5.8) and (5.9) is simplicity and uniformity — every mesh point is treated as an "unknown," and the boundary conditions are just special equations in the system.

Let $m_1, m_2 \geq 2$ be integers. Assume the discrete points

$$0 = s_0 < s_1 < \cdots < s_{m_1} = S_{\max}, \tag{5.10}$$
$$0 = v_0 < v_1 < \cdots < v_{m_2} = V_{\max}. \tag{5.11}$$

So the interior points are those (s_i, v_j) with $1 \leq i \leq m_1 - 1$ and $1 \leq j \leq m_2 - 1$. The boundary points are those (s_i, v_j) with $i \in \{0, m_1\}$ and $j \in \{0, m_2\}$. Let

$$U_{ij}(t) \coloneqq U(t, s_i, v_j), \tag{5.12}$$
$$\alpha^0_{ij}(t) \coloneqq \alpha_0(t, s_i, v_j), \tag{5.13}$$
$$\alpha^1_{ij}(t) \coloneqq \alpha_1(t, s_i, v_j), \tag{5.14}$$
$$\alpha^2_{ij}(t) \coloneqq \alpha_2(t, s_i, v_j), \tag{5.15}$$
$$\beta^{11}_{ij}(t) \coloneqq \beta_{11}(t, s_i, v_j), \tag{5.16}$$
$$\beta^{12}_{ij}(t) \coloneqq \beta_{12}(t, s_i, v_j), \tag{5.17}$$
$$\beta^{22}_{ij}(t) \coloneqq \beta_{22}(t, s_i, v_j). \tag{5.18}$$

The semidiscretized version of (5.1) is

$$U'_{ij}(t) = \alpha^0_{ij}(t) U_{i,j}(t) + \alpha^1_{ij}(t) \sum_{k=-1}^{1} \omega^1_{i,k} U_{i+k,j}(t) + \alpha^2_{ij}(t) \sum_{k=-1}^{1} \omega^2_{j,k} U_{i,j+k}(t) \tag{5.19}$$
$$+ \beta^{11}_{ij}(t) \sum_{k=-1}^{1} \omega^{11}_{i,k} U_{i+k,j}(t) + \beta^{22}_{ij}(t) \sum_{k=-1}^{1} \omega^{22}_{j,k} U_{i,j+k}(t) \tag{5.20}$$
$$+ 2\beta^{12}_{ij}(t) \sum_{k_1,k_2=-1}^{1} \omega^1_{i,k_1} \omega^2_{j,k_2} U_{i+k_1,j+k_2}(t). \tag{5.21}$$

Let $\mathrm{D}_s, \mathrm{D}_{ss} \in \mathbb{R}^{(m_1+1)\times(m_1+1)}$ and $\mathrm{D}_v, \mathrm{D}_{vv} \in \mathbb{R}^{(m_2+1)\times(m_2+1)}$ such that

$$\mathrm{D}_s = \begin{bmatrix} \omega^1_{0,0} & \omega^1_{0,1} & \omega^1_{0,2} & & & \\ \omega^1_{1,-1} & \omega^1_{1,0} & \omega^1_{1,1} & & & \\ & \omega^1_{2,-1} & \omega^1_{2,0} & \omega^1_{2,1} & & \\ & & \ddots & \ddots & \ddots & \\ & & & \omega^1_{m_1-1,-1} & \omega^1_{m_1-1,0} & \omega^1_{m_1-1,1} \\ & & & \omega^1_{m_1,-2} & \omega^1_{m_1,-1} & \omega^1_{m_1,0} \end{bmatrix}, \tag{5.22}$$

$$\mathrm{D}_v = \begin{bmatrix} \omega^2_{0,0} & \omega^2_{0,1} & \omega^2_{0,2} & & & \\ \omega^2_{1,-1} & \omega^2_{1,0} & \omega^2_{1,1} & & & \\ & \omega^2_{2,-1} & \omega^2_{2,0} & \omega^2_{2,1} & & \\ & & \ddots & \ddots & \ddots & \\ & & & \omega^2_{m_2-1,-1} & \omega^2_{m_2-1,0} & \omega^2_{m_2-1,1} \\ & & & \omega^2_{m_2,-2} & \omega^2_{m_2,-1} & \omega^2_{m_2,0} \end{bmatrix}, \tag{5.23}$$

$$\mathrm{D}_{ss} = \begin{bmatrix} \omega^{11}_{0,0} & \omega^{11}_{0,1} & \omega^{11}_{0,2} & & & \\ \omega^{11}_{1,-1} & \omega^{11}_{1,0} & \omega^{11}_{1,1} & & & \\ & \omega^{11}_{2,-1} & \omega^{11}_{2,0} & \omega^{11}_{2,1} & & \\ & & \ddots & \ddots & \ddots & \\ & & & \omega^{11}_{m_1-1,-1} & \omega^{11}_{m_1-1,0} & \omega^{11}_{m_1-1,1} \\ & & & \omega^{11}_{m_1,-2} & \omega^{11}_{m_1,-1} & \omega^{11}_{m_1,0} \end{bmatrix}, \tag{5.24}$$

$$\mathrm{D}_{vv} = \begin{bmatrix} \omega^{22}_{0,0} & \omega^{22}_{0,1} & \omega^{22}_{0,2} & & & \\ \omega^{22}_{1,-1} & \omega^{22}_{1,0} & \omega^{22}_{1,1} & & & \\ & \omega^{22}_{2,-1} & \omega^{22}_{2,0} & \omega^{22}_{2,1} & & \\ & & \ddots & \ddots & \ddots & \\ & & & \omega^{22}_{m_2-1,-1} & \omega^{22}_{m_2-1,0} & \omega^{22}_{m_2-1,1} \\ & & & \omega^{22}_{m_2,-2} & \omega^{22}_{m_2,-1} & \omega^{22}_{m_2,0} \end{bmatrix}. \tag{5.25}$$

We use

- Top row: forward difference.
- Bottom row: backward difference.
- Interior rows: central difference.

In 't Hout's implementation (slide[3] 40) $\mathrm{D}_s, \mathrm{D}_{ss}, \mathrm{D}_v, \mathrm{D}_{vv}$ are constructed as submatrices restricted to interior indices, which requires careful bookkeeping to separate

[3] http://quantshub.com/qhworkshopview/22

known boundary values from interior unknowns. Our formulation avoids this by treating the entire grid uniformly. As such, our approach to dealing with boundary conditions is more straighforward to implement on computers.

Denote by $U(t)$ the matrix of all entries $U_{ij}(t)$ for $0 \le i \le m_1$ and $0 \le j \le m_2$. Then $U(t) \in \mathbb{R}^{(m_1+1)\times(m_2+1)}$.

Let $\Omega^0(t), \Omega^1(t), \Omega^2(t), \Omega^{11}(t), \Omega^{22}(t), \Omega^{12}(t) \in \mathbb{R}^{(m_1+1)\times(m_2+1)}$ such that

$$\Omega^0_{ij}(t) = \alpha^0_{ij}(t), \qquad \Omega^1_{ij}(t) = \alpha^1_{ij}(t), \qquad \Omega^2_{ij}(t) = \alpha^2_{ij}(t), \tag{5.26}$$

$$\Omega^{11}_{ij}(t) = \beta^{11}_{ij}(t), \qquad \Omega^{22}_{ij}(t) = \beta^{22}_{ij}(t), \qquad \Omega^{12}_{ij}(t) = \beta^{12}_{ij}(t). \tag{5.27}$$

Denote by $\odot$ the Hadamard (pointwise) product. Then

$$U'(t) = \Omega^0(t) \odot U(t) + \Omega^1(t) \odot [\mathrm{D}_s U(t)] + \Omega^2(t) \odot [U(t)\mathrm{D}_v^\top] \tag{5.28}$$

$$+ \Omega^{11}(t) \odot [\mathrm{D}_{ss} U(t)] + \Omega^{22}(t) \odot [U(t)\mathrm{D}_{vv}^\top] \tag{5.29}$$

$$+ 2\Omega^{12}(t) \odot [\mathrm{D}_s U(t)\mathrm{D}_v^\top]. \tag{5.30}$$

At the boundary part corresponding to $v = 0$, we discretize $\partial_v u(t, s, 0)$ and $\partial^2_{vv} u(t, s, 0)$ using forward difference scheme.

5.1 Incorporating (5.8) and (5.9)

To enforce (5.8) at $s = 0$ (top row) and (5.9) at $v = V_{\max}$ (last column), we nullify the corresponding rows and columns of $\Omega^0(t), \Omega^1(t), \Omega^2(t), \Omega^{11}(t), \Omega^{22}(t), \Omega^{12}(t)$: for all $0 \le i \le m_1$ and $0 \le j \le m_2$,

$$\Omega^0_{0j}(t) = \Omega^0_{i,m_2}(t) := 0, \tag{5.31}$$

$$\Omega^1_{0j}(t) = \Omega^1_{i,m_2}(t) := 0, \tag{5.32}$$

$$\Omega^2_{0j}(t) = \Omega^2_{i,m_2}(t) := 0, \tag{5.33}$$

$$\Omega^{11}_{0j}(t) = \Omega^{11}_{i,m_2}(t) := 0, \tag{5.34}$$

$$\Omega^{22}_{0j}(t) = \Omega^{22}_{i,m_2}(t) := 0, \tag{5.35}$$

$$\Omega^{12}_{0j}(t) = \Omega^{12}_{i,m_2}(t) := 0. \tag{5.36}$$

We compensate by constructing a forcing term $G^1(t) \in \mathbb{R}^{(m_1+1)\times(m_2+1)}$: for all $0 \le i \le m_1$,

$$G^1_{i,m_2}(t) := -r_f s_i e^{-r_f t}. \tag{5.37}$$

This is analogous to the matrix G on slide[4] 38 of in 't Hout, which encodes the

[4] http://quantshub.com/qhworkshopview/22

boundary value $u(t, s, V_{\max}) = se^{-r_f t}$ as a correction to the right-hand side. The difference is that our G^1 arises from enforcing the time-derivative condition (5.9) rather than prescribing the value directly.

As such, we have avoided the discretization of spatial derivatives at the boundary parts corresponding to $s = 0$ and to $v = V_{\max}$.

5.2 Incorporating (5.7)

We nullify the bottom row of D_s: for all $0 \leq j \leq m_1$,

$$[\mathrm{D}_s]_{m_1,j} := 0. \tag{5.38}$$

We compensate by constructing a matrix $E^1(t) \in \mathbb{R}^{(m_1+1)\times(m_2+1)}$: for all $0 \leq j \leq m_2$,

$$E^1_{m_1,j}(t) := e^{-r_f t}. \tag{5.39}$$

We also use (5.7) to eliminate the ghost point s_{m_1+1} when computing $\partial^2_{ss} u(t, S_{\max}, v)$ as follows. We have $f_{i+1} - f_{i-1} \approx f'_i(\Delta_i + \Delta_{i+1})$, so $f_{i+1} \approx f_{i-1} + f'_i(\Delta_i + \Delta_{i+1})$. Then

$$\begin{aligned} f''_i &\approx \delta_{-1} f_{i-1} + \delta_0 f_i + \delta_1 f_{i+1} && (5.40)\\ &\approx \delta_{-1} f_{i-1} + \delta_0 f_i + \delta_1 [f_{i-1} + f'_i(\Delta_i + \Delta_{i+1})] && (5.41)\\ &= (\delta_{-1} + \delta_1) f_{i-1} + \delta_0 f_i + \delta_1 (\Delta_i + \Delta_{i+1}) f'_i. && (5.42) \end{aligned}$$

Thus we nullify the bottom row of D_{ss} and re-define

$$\begin{aligned} [\mathrm{D}_{ss}]_{m_1,m_1-2} &:= 0, && (5.43)\\ [\mathrm{D}_{ss}]_{m_1,m_1-1} &:= \delta_{-1} + \delta_1, && (5.44)\\ [\mathrm{D}_{ss}]_{m_1,m_1} &:= \delta_0. && (5.45) \end{aligned}$$

We supplement by constructing a matrix $E^{11}(t) \in \mathbb{R}^{(m_1+1)\times(m_2+1)}$: for all $0 \leq j \leq m_2$,

$$E^{11}_{m_1,j}(t) := \delta_1(\Delta_{m_1} + \Delta_{m_1+1}) e^{-r_f t}. \tag{5.46}$$

As such, we have avoided the discretization of $\partial_s u(t, S_{\max}, v)$ and the ghost point in discretizing $\partial^2_{ss} u(t, S_{\max}, v)$.

At this point, every grid point — boundary or interior — is treated as an unknown governed by a uniform ODE, with the boundary conditions absorbed into the modified matrices and forcing terms.

5.3 The matrix-valued linear ODE

This section collects the modifications from Sections 5.1 and 5.2 as an intermediate step before vectorizing in Section 5.4. Recall from above treatment that we have created new matrices $G^1(t), E^1(t), E^{11}(t)$ and modified

- the top row and last column of $\Omega^0(t), \Omega^1(t), \Omega^2(t), \Omega^{11}(t), \Omega^{22}(t), \Omega^{12}(t)$.
- the bottom row of $\mathrm{D}_s, \mathrm{D}_{ss}$.

Thus we arrive at

$$\begin{aligned} U'(t) &= \Omega^0(t) \odot U(t) + \Omega^1(t) \odot [\mathrm{D}_s U(t) + E^1(t)] + \Omega^2(t) \odot [U(t)\mathrm{D}_v^\top] \\ &\quad + \Omega^{11}(t) \odot [\mathrm{D}_{ss} U(t) + E^{11}(t)] + \Omega^{22}(t) \odot [U(t)\mathrm{D}_{vv}^\top] \\ &\quad + 2\Omega^{12}(t) \odot [\mathrm{D}_s U(t)\mathrm{D}_v^\top] + G^1(t) \\ &= \Omega^0(t) \odot U(t) + \Omega^1(t) \odot [\mathrm{D}_s U(t)] + \Omega^2(t) \odot [U(t)\mathrm{D}_v^\top] \\ &\quad + \Omega^{11}(t) \odot [\mathrm{D}_{ss} U(t)] + \Omega^{22}(t) \odot [U(t)\mathrm{D}_{vv}^\top] \\ &\quad + 2\Omega^{12}(t) \odot [\mathrm{D}_s U(t)\mathrm{D}_v^\top] \\ &\quad + G^1(t) + \Omega^1(t) \odot E^1(t) + \Omega^{11}(t) \odot E^{11}(t). \end{aligned} \tag{5.47}$$

(5.48)

5.4 The vector-valued linear ODE

We denote by vec the operator that turns a matrix into a vector by putting its successive columns below each other. Let $\otimes$ denote the Kronecker product. We will use the following result to transform (5.48).

Lemma 5.1. *1. Let X, W, Y be matrices of sizes $m_1 \times m_1, m_1 \times m_2, m_2 \times m_2$. Then*

$$\mathrm{vec}[XWY^\top] = [Y \otimes X]\,\mathrm{vec}[W].$$

2. Let W_1, W_2 be matrices of size $m_1 \times m_2$. Then

$$\mathrm{vec}[W_1 \odot W_2] = \mathrm{vec}[W_1] \odot \mathrm{vec}[W_2].$$

3. Let W_1, W_2 be two vectors of the same size. Then

$$W_1 \odot W_2 = \mathrm{diag}[W_1]W_2.$$

By Lemma 5.1,

$$\begin{aligned}
\mathrm{vec}[\Omega^1(t) \odot [\mathrm{D}_s U(t)]] &= \mathrm{vec}[\Omega^1(t)] \odot \mathrm{vec}[\mathrm{D}_s U(t) \mathrm{I}_{m_2}] && (5.49)\\
&= \mathrm{vec}[\Omega^1(t)] \odot [[\mathrm{I}_{m_2} \otimes \mathrm{D}_s] \,\mathrm{vec}[U(t)]] && (5.50)\\
&= \mathrm{diag}[\mathrm{vec}[\Omega^1(t)]][\mathrm{I}_{m_2} \otimes \mathrm{D}_s] \,\mathrm{vec}[U(t)]. && (5.51)
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathrm{vec}[\Omega^0(t) \odot U(t)] &= \mathrm{diag}[\mathrm{vec}[\Omega^0(t)]][\mathrm{I}_{m_2} \otimes \mathrm{I}_{m_1}] \,\mathrm{vec}[U(t)], && (5.52)\\
\mathrm{vec}[\Omega^2(t) \odot [U(t)\mathrm{D}_v^\top]] &= \mathrm{diag}[\mathrm{vec}[\Omega^2(t)]][\mathrm{D}_v \otimes \mathrm{I}_{m_1}] \,\mathrm{vec}[U(t)], && (5.53)\\
\mathrm{vec}[\Omega^{11}(t) \odot [\mathrm{D}_{ss} U(t)]] &= \mathrm{diag}[\mathrm{vec}[\Omega^{11}(t)]][\mathrm{I}_{m_2} \otimes \mathrm{D}_{ss}] \,\mathrm{vec}[U(t)], && (5.54)\\
\mathrm{vec}[\Omega^{22}(t) \odot [U(t)\mathrm{D}_{vv}^\top]] &= \mathrm{diag}[\mathrm{vec}[\Omega^{22}(t)]][\mathrm{D}_{vv} \otimes \mathrm{I}_{m_1}] \,\mathrm{vec}[U(t)], && (5.55)\\
\mathrm{vec}[\Omega^{12}(t) \odot [\mathrm{D}_s U(t)\mathrm{D}_v^\top]] &= \mathrm{diag}[\mathrm{vec}[\Omega^{12}(t)]][\mathrm{D}_v \otimes \mathrm{D}_s] \,\mathrm{vec}[U(t)]. && (5.56)
\end{aligned}$$

Let

$$\begin{aligned}
\boldsymbol{U}(t) &\coloneqq \mathrm{vec}[U(t)], && (5.57)\\
A^0(t) &\coloneqq 2\,\mathrm{diag}[\mathrm{vec}[\Omega^{12}(t)]][\mathrm{D}_v \otimes \mathrm{D}_s], && (5.58)\\
A^1(t) &\coloneqq \mathrm{diag}[\mathrm{vec}[\Omega^1(t)]][\mathrm{I}_{m_2} \otimes \mathrm{D}_s] + \mathrm{diag}[\mathrm{vec}[\Omega^{11}(t)]][\mathrm{I}_{m_2} \otimes \mathrm{D}_{ss}] && (5.59)\\
&\quad + \frac{1}{2}\,\mathrm{diag}[\mathrm{vec}[\Omega^0(t)]][\mathrm{I}_{m_2} \otimes \mathrm{I}_{m_1}],\\
A^2(t) &\coloneqq \mathrm{diag}[\mathrm{vec}[\Omega^2(t)]][\mathrm{D}_v \otimes \mathrm{I}_{m_1}] + \mathrm{diag}[\mathrm{vec}[\Omega^{22}(t)]][\mathrm{D}_{vv} \otimes \mathrm{I}_{m_1}] && (5.60)\\
&\quad + \frac{1}{2}\,\mathrm{diag}[\mathrm{vec}[\Omega^0(t)]][\mathrm{I}_{m_2} \otimes \mathrm{I}_{m_1}],\\
g_0(t) &\coloneqq 0, && (5.61)\\
g_1(t) &\coloneqq \mathrm{vec}[\frac{1}{2}G^1(t) + \Omega^1(t) \odot E^1(t) + \Omega^{11}(t) \odot E^{11}(t)], && (5.62)\\
g_2(t) &\coloneqq \mathrm{vec}[\frac{1}{2}G^1(t)]. && (5.63)
\end{aligned}$$

We choose the matrix A^0 as the part that stems from the discretization of the mixed derivative term in (5.1). Next, in line with the classical ADI idea, A^1 and

A^2 are the two parts that correspond to all spatial derivatives in the s- and v-directions, respectively. The term α_0 in (5.1) is distributed evenly over A^1, A^2. The construction of the forcing terms (g_0, g_1, g_2) is analogous to that of (A^0, A^1, A^2).

Then (5.48) becomes

$$\boldsymbol{U}'(t) = [A^0(t) + A^1(t) + A^2(t)]\boldsymbol{U}(t) + [g_0(t) + g_1(t) + g_2(t)].$$

Notice that

- $\boldsymbol{U}'(t), \boldsymbol{U}(t), g_0(t), g_1(t), g_2(t)$ are column vectors of size $(m_1+1)(m_2+1)$.
- $A^0(t), A^1(t), A^2(t)$ are matrices of size $(m_1+1)(m_2+1)\times(m_1+1)(m_2+1)$.

6 Alternating Direction Implicit method

Consider the vector-valued ODE

$$\begin{cases} \boldsymbol{U}'(t) &= A(t)\boldsymbol{U}(t) + g(t) \quad \text{for} \quad t \in [0, T], \\ \boldsymbol{U}(0) &= \varphi, \end{cases}$$

where $\boldsymbol{U}(t), g(t)$ are of size m, $A(t)$ is of size $m \times m$, and

$$A(t) = A^0(t) + A^1(t) + A^2(t), \tag{6.1}$$
$$g(t) = g_0(t) + g_1(t) + g_2(t). \tag{6.2}$$

Above,

- A^0 corresponds to $\partial^2_{sv}u$ term.
- A^1 corresponds to $\partial_s u, \partial^2_{ss}u$ term.
- A^2 corresponds to $\partial_v u, \partial^2_{vv}u$ term.

Analogous correspondence for $(g_0(t), g_1(t), g_2(t))$ holds.

For $t \in [0, T]$ and $w \in \mathbb{R}^m$, Let

$$F(t, w) := A(t)w + g(t), \tag{6.3}$$
$$F_j(t, w) := A^j(t)w + g_j(t) \quad \text{for} \quad j = 0, 1, 2. \tag{6.4}$$

The step size $\Delta t := T/N$ with given integer $N \geq 1$ and temporal grid points $t_n := n\Delta t$.

The standard Douglas (Do) scheme is

$$\begin{cases} Y_0 &= U_{n-1} + \Delta t F(t_{n-1}, U_{n-1}), \\ Y_j &= Y_{j-1} + \theta \Delta t [F_j(t_n, Y_j) - F_j(t_{n-1}, U_{n-1})] \quad \text{for} \quad j = 1, 2, \\ U_n &= \widetilde{Y}_2, \end{cases}$$

with $\theta = \frac{1}{2}$. In Do scheme, an explicit predictor stage is followed by two implicit corrector stages. If A^0 is non-zero, then Do scheme has just order 1. We present the Douglas scheme here as a baseline; the Modified Craig-Sneyd (MCS) scheme below resolves this order reduction:

$$(6.5)\begin{cases} Y_0 = U_{n-1} + \Delta t F(t_{n-1}, U_{n-1}), & (6.5a) \\ Y_j = Y_{j-1} + \theta \Delta t [F_j(t_n, Y_j) - F_j(t_{n-1}, U_{n-1})] \quad \text{for} \quad j = 1, 2, & (6.5b) \\ \widehat{Y}_0 = Y_0 + \theta \Delta t [F_0(t_n, Y_2) - F_0(t_{n-1}, U_{n-1})], & (6.5c) \\ \widetilde{Y}_0 = \widehat{Y}_0 + \Big(\dfrac{1}{2} - \theta\Big) \Delta t [F(t_n, Y_2) - F(t_{n-1}, U_{n-1})], & (6.5d) \\ \widetilde{Y}_j = \widetilde{Y}_{j-1} + \theta \Delta t [F_j(t_n, \widetilde{Y}_j) - F_j(t_{n-1}, U_{n-1})] \quad \text{for} \quad j = 1, 2, & (6.5e) \\ U_n = \widetilde{Y}_2, & (6.5f) \end{cases}$$

with $\theta = \frac{1}{3}$.

The computation chain is

$$U_{n-1} \mapsto Y_0 \mapsto Y_1 \mapsto Y_2 \mapsto \widehat{Y}_0 \mapsto \widetilde{Y}_0 \mapsto \widetilde{Y}_1 \mapsto \widetilde{Y}_2 =: U_n.$$

MCS scheme has order 2 and can be seen as an extension of Do scheme with

- Additional explicit predictor stage and two implicit corrector stages.
- Four linear systems per time step with $\mathrm{I}_m - \theta \Delta t A^1$ and $\mathrm{I}_m - \theta \Delta t A^2$.
- The A^0 part is always treated in an explicit fashion.

The steps (6.5a), (6.5c) and (6.5d) are explicit, i.e., we just compute the right-hand side to get the value of the left-hand side. The steps (6.5b) and (6.5e) are implicit, i.e., we have to solve systems of linear equations. Next we obtain a more convenient forms of these systems.

We have (6.5b) is equivalent to

$$[\mathrm{I}_m - \theta \Delta t A^j(t_n)] Y_j = Y_{j-1} + \theta \Delta t [g_j(t_n) - F_j(t_{n-1}, U_{n-1})].$$

We have (6.5e) is equivalent to

$$[\mathrm{I}_m - \theta \Delta t A^j(t_n)]\widetilde{Y}_j = \widetilde{Y}_{j-1} + \theta \Delta t[g_j(t_n) - F_j(t_{n-1}, U_{n-1})].$$

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